Streamlit Methods

1. `st.title()`

Purpose: Show main title of the app.

```
import streamlit as st

st.title("My First Streamlit App")
```

2. `st.header()`

Purpose: Section heading.

```
st.header("User Information")
```

3. `st.subheader()`

Purpose: Sub-section heading.

```
st.subheader("Personal Details")
```

4. `st.text()`

Purpose: Display plain text (no formatting).

```
st.text("This is simple text")
```

⚠️ No markdown, no styling

5. `st.write()`

Purpose: Most flexible function (text, numbers, lists, dicts, DataFrames).

```
st.write("Hello Streamlit")
st.write(123)
st.write([1, 2, 3])
st.write({"name": "Sumit", "role": "Trainer"})
```

👉 **Most used Streamlit function**

6. `st.markdown()`

Purpose: Display formatted text using Markdown.

```
st.markdown("### This is Markdown")
st.markdown("**Bold Text**")
st.markdown("*Italic Text*")
st.markdown("- Item 1\n- Item 2")
```

Advanced (HTML allowed):

```
st.markdown("<h3 style='color:red'>Red Text</h3>",
unsafe_allow_html=True)
```

7. `st.caption()`

Purpose: Small helper text (notes, hints).

```
st.caption("This is a caption text")
```

Used for explanations under charts or inputs.

8. `st.code()`

Purpose: Show code snippets.

```
st.code("""
def add(a, b):
    return a + b
""", language="python")
```

Languages supported: python, sql, json, etc.

9. `st.latex()`

Purpose: Display mathematical equations (LaTeX).

```
st.latex(r'''
a^2 + b^2 = c^2
''')
```

Used in **ML / Math / AI explanations**

10. `st.divider()`

Purpose: Visual horizontal line separator.

```
st.divider()
```

Helps structure UI sections.

Mini Example (using first 10 together)

```
import streamlit as st

st.title("Student Dashboard")
st.header("Profile")
st.subheader("Basic Info")

st.text("Name: Sumit")
st.write("Age:", 25)

st.markdown("**Skills:** Python, ML, Streamlit")
st.caption("Data updated today")

st.code("print('Hello Streamlit')", language="python")

st.latex(r"E = mc^2")

st.divider()
st.write("End of Section")
```

1. `st.button()`

Purpose: Trigger an action on click.

```
import streamlit as st

if st.button("Click Me"):
```

```
    st.write("Button Clicked!")
```

- Returns True when clicked

12. st.text_input()

Purpose: Take single-line text input.

```
name = st.text_input("Enter your name")
st.write("Name:", name)
```

With default value:

```
st.text_input("City", value="Pune")
```

13. st.number_input()

Purpose: Take numeric input.

```
age = st.number_input("Enter age", min_value=0, max_value=100)
st.write("Age:", age)
```

Float input:

```
salary = st.number_input("Salary", step=1000.0)
```

14. st.text_area()

Purpose: Multi-line text input.

```
feedback = st.text_area("Enter feedback")
st.write(feedback)
```

Used for:

- Comments
- Address
- Description

15. st.checkbox()

Purpose: True / False input.

```
agree = st.checkbox("I agree to terms")

if agree:
    st.write("Thank you for agreeing")
```

16. st.radio()

Purpose: Select **one option**.

```
gender = st.radio(
    "Select Gender",
    ["Male", "Female", "Other"]
)
st.write("Selected:", gender)
```

17. st.selectbox()

Purpose: Dropdown (single select).

```
course = st.selectbox(
    "Choose course",
    ["Python", "ML", "DL", "AI"]
)
st.write("Course:", course)
```

18. st.multiselect()

Purpose: Select **multiple options**.

```
skills = st.multiselect(
    "Select skills",
    ["Python", "SQL", "ML", "Streamlit"]
)
st.write("Skills:", skills)
```

19. st.slider()

Purpose: Select value from a range.

```
rating = st.slider("Rate us", 1, 5)
st.write("Rating:", rating)
```

Range slider:

```
age_range = st.slider("Age range", 18, 60, (20, 30))
```

20. st.file_uploader()

Purpose: Upload files.

```
file = st.file_uploader("Upload a CSV file")

if file is not None:
    st.write("File name:", file.name)
```

Read CSV:

```
import pandas as pd

df = pd.read_csv(file)
st.write(df)
```

Mini Example (Form-like UI)

```
import streamlit as st

st.title("Student Registration")

name = st.text_input("Name")
age = st.number_input("Age", 0, 100)
gender = st.radio("Gender", ["Male", "Female"])
skills = st.multiselect("Skills", ["Python", "ML", "SQL"])
agree = st.checkbox("I agree")

if st.button("Submit"):
    st.write("Name:", name)
    st.write("Age:", age)
    st.write("Gender:", gender)
    st.write("Skills:", skills)
```

21. st.form()

Purpose: Group inputs and submit together (used in CRUD apps).

```python
import streamlit as st

with st.form("my_form"):
    name = st.text_input("Name")
    age = st.number_input("Age", 0, 100)
    submit = st.form_submit_button("Submit")

if submit:
    st.write(name, age)
```

✅ Inputs run only after submit

22. `st.form_submit_button()`

Purpose: Submit button inside a form.

```python
with st.form("login"):
    user = st.text_input("Username")
    pwd = st.text_input("Password", type="password")
    login = st.form_submit_button("Login")

if login:
    st.write("Login Successful")
```

23. `st.columns()`

Purpose: Create side-by-side layout.

```python
col1, col2 = st.columns(2)

with col1:
    st.text_input("First Name")

with col2:
```

```python
    st.text_input("Last Name")
```

3 columns:

```python
c1, c2, c3 = st.columns(3)
```

24. `st.container()`

Purpose: Group elements dynamically.

```python
container = st.container()

container.write("Inside container")
container.button("Click")
```

Used for:

- Dynamic updates
- Loops

25. `st.expander()`

Purpose: Collapsible section.

```python
with st.expander("See details"):
    st.write("Hidden content")
```

Used for:

- Advanced info
- Logs
- Help text

26. `st.sidebar`

Purpose: Sidebar navigation / filters.

```
st.sidebar.title("Menu")
option = st.sidebar.selectbox(
    "Choose page",
    ["Home", "About", "Contact"]
)
```

Sidebar input:

```
age = st.sidebar.slider("Age", 18, 60)
```

27. `st.success()`

Purpose: Success message.

```
st.success("Data saved successfully")
```

Used after:

- Insert
- Update
- Submit

28. `st.error()`

Purpose: Error message.

```
st.error("Something went wrong")
```

Used in:

- Exceptions
- Validation errors

29. `st.warning()`

Purpose: Warning message.

```
st.warning("Please fill all fields")
```

30. `st.info()`

Purpose: Informational message.

```
st.info("This is an information message")
```

Mini CRUD-style Example

```
import streamlit as st

st.title("Student Form")

with st.form("student_form"):
    name = st.text_input("Name")
    age = st.number_input("Age", 0, 100)
    submit = st.form_submit_button("Save")

if submit:
    if name == "":
        st.warning("Name required")
    else:
        st.success("Student saved")
```

31. `st.dataframe()`

Purpose: Display interactive DataFrame.

```
import streamlit as st
import pandas as pd

df = pd.DataFrame({
    "Name": ["A", "B"],
    "Marks": [80, 90]
})

st.dataframe(df)
```

✅ Sortable, scrollable

32. `st.table()`

Purpose: Display static table.

```
st.table(df)
```

⚠️ No sorting or scrolling

33. `st.metric()`

Purpose: Show KPIs (numbers + delta).

```
st.metric("Total Students", 120, "+10")
```

Used in dashboards.

34. `st.json()`

Purpose: Display JSON data nicely.

```
data = {
    "name": "Sumit",
    "role": "Trainer",
    "skills": ["Python", "ML"]
}

st.json(data)
```

35. `st.image()`

Purpose: Display images.

```
st.image("image.jpg", caption="Sample Image")
```

From URL:

```
st.image("https://example.com/img.png")
```

36. `st.audio()`

Purpose: Play audio.

```
st.audio("audio.mp3")
```

37. `st.video()`

Purpose: Play video.

```
st.video("video.mp4")
```

YouTube:

```
st.video("https://youtu.be/xxxx")
```

38. `st.pyplot()`

Purpose: Display Matplotlib plots.

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots()
ax.plot([1, 2, 3], [10, 20, 30])

st.pyplot(fig)
```

Used for:

- ML plots
- Custom charts

39. `st.line_chart()`

Purpose: Quick line chart.

```
import pandas as pd

df = pd.DataFrame({
    "sales": [10, 20, 30, 40]
```

```
})

st.line_chart(df)
```

40. `st.bar_chart()`

Purpose: Quick bar chart.

```
st.bar_chart(df)
```

Mini Dashboard Example

```
import streamlit as st
import pandas as pd

st.title("Dashboard")

df = pd.DataFrame({
    "Month": ["Jan", "Feb", "Mar"],
    "Sales": [100, 150, 200]
})

st.metric("Total Sales", sum(df["Sales"]))
st.dataframe(df)
st.line_chart(df["Sales"])
```

41. `st.session_state`

Purpose: Store data across user interactions.

```
import streamlit as st
```

```
if "count" not in st.session_state:
    st.session_state.count = 0

if st.button("Increase"):
    st.session_state.count += 1

st.write("Count:", st.session_state.count)
```

✅ Used for:

- Login state
- Counters
- Multi-step forms

42. `st.cache_data`

Purpose: Cache data (fast performance).

```
@st.cache_data
def load_data():
    return [1, 2, 3, 4]

data = load_data()
st.write(data)
```

Used for:

- Database reads
- CSV loading
- API calls

43. `st.cache_resource`

Purpose: Cache heavy resources.

```
@st.cache_resource
def load_model():
    return "ML Model Loaded"

model = load_model()
```

Used for:

- ML models
- DB connections

44. `st.spinner()`

Purpose: Show loading indicator.

```
import time

with st.spinner("Loading..."):
    time.sleep(3)

st.success("Done!")
```

45. `st.progress()`

Purpose: Progress bar.

```
import time

progress = st.progress(0)
```

```
for i in range(100):
    time.sleep(0.05)
    progress.progress(i + 1)
```

46. st.empty()

Purpose: Placeholder for dynamic content.

```
placeholder = st.empty()

placeholder.text("Loading...")
placeholder.success("Finished")
```

47. st.stop()

Purpose: Stop execution.

```
name = st.text_input("Name")

if name == "":
    st.warning("Enter name")
    st.stop()

st.write("Hello", name)
```

48. st.rerun()

Purpose: Rerun the app manually.

```
if st.button("Refresh"):
    st.rerun()
```

49. st.query_params

Purpose: Read URL parameters.

```
st.write(st.query_params)
```

Example URL:

http://localhost:8501/?user=admin

50. st.set_page_config()

Purpose: Configure page settings.

```
st.set_page_config(
    page_title="My App",
    layout="wide",
    initial_sidebar_state="expanded"
)
```